

# **Thermodynamics – Assignment 05**

Instructor: Dr. Michael Caracotsios

University of Illinois at Chicago

Name: Yahir Cordova

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## Abstract

In this report, two thermodynamic problems involving vapor–liquid equilibrium and gas solubility are analyzed using fundamental principles from phase equilibrium and mixture thermodynamics.

In Problem A, a multi-component hydrocarbon mixture undergoes a pressure reduction across a throttling valve, resulting in partial vaporization. The system is modeled using Raoult’s law to relate liquid and vapor phase compositions, and the Rachford–Rice equation is applied to determine the vapor fraction. Since the outlet temperature is not specified, the excel solver is implemented in which the equilibrium relations are coupled with an isenthalpic energy balance to determine both the final temperature and phase compositions instantaneously.

In Problem B, the solubility of oxygen in water at ambient conditions is evaluated using Henry’s law, which is appropriate for supercritical gas components dissolved in liquid solutions. The effect of water vapor in the gas phase is accounted for by correcting the oxygen partial pressure based on relative humidity. The resulting liquid-phase mole fraction is then converted to a mass concentration in mg/L, providing a physically meaningful measure of dissolved oxygen.

Together, these problems demonstrate the application of thermodynamic models for phase equilibrium, including Raoult’s law, Henry’s law, and flash calculations, as well as the integration of equilibrium relations with energy balances to solve practical engineering problems.

## 1 Problem A: Background

In this problem, a multi-component hydrocarbon mixture undergoes a pressure reduction across a throttling valve. The inlet stream is specified at a temperature of  $0^{\circ}\text{C}$  and a pressure of 20 bar, with an overall composition given by:

$$z = [0.2; 0.5; 0.3]$$

corresponding to ethylene, propane, and iso-butane, respectively.

The mixture is expanded through a valve to a final pressure of 1 bar. The throttling process is assumed to be adiabatic with no shaft work, which implies that the enthalpy of the system remains constant across the valve. As a result of the pressure drop, part of the liquid mixture vaporizes, leading to a two-phase system at the outlet consisting of both liquid and vapor.

The vapor and liquid phases are assumed to behave as an ideal gas mixture and an ideal liquid solution, respectively. Under these assumptions, Raoult’s law is applicable for relating the phase compositions.

Because the outlet temperature is not specified, it must be determined as part of the solution. Additionally, the extent of vaporization, represented by the vapor fraction, is also unknown. These quantities must be determined simultaneously while satisfying both phase equilibrium and the energy balance for the system.

The objective of this problem is to determine the outlet temperature, the vapor fraction, and the compositions of the liquid and vapor phases after the throttling process.

### 1.1 Dew Point and Bubble Point Temperature Calculations - Q1

First, the three components of the mixture were added to the CHECalculator, allowing for direct access to thermodynamic properties such as saturation pressure.

A trial temperature, labeled as the dew point temperature in degrees Celsius, was defined in a designated cell. Using this temperature, the saturation pressure of each component was calculated in separate cells using the **Psat()** function, with each function referencing the corresponding component and the trial temperature cell. At this stage, the calculated saturation pressures are based on an initial temperature guess and do not yet satisfy the dew point condition.

Next, the  $K$ -values were calculated for each component using:

$$K_i = \frac{P_i^{\text{sat}}(T)}{P}$$

These values were then used to evaluate the dew point expression, implemented in a separate cell as:

$$\sum_i \frac{z_i}{K_i}$$

Because the saturation pressures and  $K$ -values are functions of the temperature guess, the dew point expression is implicitly dependent on this temperature. Therefore, a numerical method was required to determine the correct value. Goal Seek was applied by setting the cell containing the dew point expression equal to 1, while varying the cell containing the temperature guess. The **dew point temperature** was determined to be 72.80°C at the inlet conditions and -31.00°C at the outlet conditions. Figure 1, below shows a summary of the results after the following numerical method was utilized.

| User Data and Calculations | Inlet          | Outlet          |
|----------------------------|----------------|-----------------|
| z1                         | 0.2000         | 0.2000          |
| z2                         | 0.5000         | 0.5000          |
| z3                         | 0.3000         | 0.3000          |
|                            |                |                 |
| P1Sat, bar                 | 119.4498       | 17.7445         |
| P2Sat, bar                 | 24.1669        | 1.5917          |
| P3Sat, bar                 | 10.8553        | 0.4447          |
|                            |                |                 |
| k1                         | 5.9725         | 17.7445         |
| k2                         | 1.2083         | 1.5917          |
| k3                         | 0.5428         | 0.4447          |
|                            |                |                 |
| Dew Point Temp, C          | <b>72.8031</b> | <b>-31.0023</b> |
| Dew Point Equation         | 1.0000         | 1.0000          |

Figure 1: Dew Point Temperature Summary

A procedure similar to that used for the dew point temperature was applied to determine the bubble point temperature. A trial temperature, labeled as the bubble point temperature in degrees Celsius, was defined in a designated cell.

Using this temperature, the saturation pressures of each component were calculated in separate cells using the **Psat()** function, with each function referencing the corresponding component and the temperature cell. These values were then used to compute updated  $K$ -values according to:

$$K_i = \frac{P_i^{\text{sat}}(T)}{P}$$

A separate cell was then used to evaluate the bubble point expression:

$$\sum_i z_i K_i$$

Because the saturation pressures and  $K$ -values depend on the temperature guess, this expression is implicitly a function of temperature. Goal Seek was applied by setting the cell containing the bubble point expression equal to 1, while varying the cell containing the bubble point temperature.

The bubble point temperature was determined to be  $32.75^\circ\text{C}$  at the inlet conditions and  $-73.37^\circ\text{C}$  at the outlet conditions. Figure 2, below shows a summary of the results after the following numerical method was utilized.

| *z-values remain the same | Inlet   | Outlet   |
|---------------------------|---------|----------|
| P1Sat, bar                | 66.3968 | 4.4536   |
| P2Sat, bar                | 10.8649 | 0.1964   |
| P3Sat, bar                | 4.2940  | 0.0370   |
|                           |         |          |
| k1                        | 3.3198  | 4.4536   |
| k2                        | 0.5432  | 0.1964   |
| k3                        | 0.2147  | 0.0370   |
|                           |         |          |
| Bubble Point Temp, C      | 32.7491 | -73.3663 |
| Bubble Point Equation     | 1.0000  | 1.0000   |

Figure 2: Bubble Point Temperature Summary

|                                               | Inlet | Outlet |
|-----------------------------------------------|-------|--------|
| Dew Point Temperature ( $^\circ\text{C}$ )    | 72.80 | -31.00 |
| Bubble Point Temperature ( $^\circ\text{C}$ ) | 32.75 | -73.37 |

Table 1: Dew and bubble point temperatures at inlet and outlet conditions

From the calculated phase equilibrium temperatures, the inlet stream is identified as a subcooled liquid since the inlet temperature is below the corresponding bubble point temperature. Therefore, no vapor is present at the inlet and the vapor fraction is zero. In contrast, the outlet temperature has not yet been determined, so the phase composition of the outlet stream remains unknown at this stage.

## 1.2 Temperature and Vapor Fraction of the Outlet Valve Q2+Q3

The analysis is then focused on the outlet stream of the process. At the outlet pressure, the  $K$ -values were recalculated; however, in this case, the saturation pressures were evaluated using a trial value for the unknown outlet temperature. The relation  $K_i = \frac{P_i^{\text{sat}}(T)}{P}$  was again used, with each saturation pressure referencing the guessed outlet temperature.

A separate cell was used to implement the Rachford–Rice equation (Eq. 5.50 in Dr.C’s textbook), which depends on the calculated  $K$ -values, the known overall composition  $z_i$ , and a trial value for the vapor fraction  $\alpha$ . Because both the  $K$ -values and the Rachford–Rice expression depend on the temperature and vapor fraction guesses, this setup allows for a numerical solution to determine the phase split at the outlet.

Because the vapor phase is assumed to behave as an ideal gas mixture and the liquid phase as an ideal liquid solution, the corresponding molar enthalpy relations from Chapter 5 were applied for each phase. Before these enthalpies could be evaluated, cells were first created for the liquid-phase and vapor-phase compositions at the outlet.

The liquid-phase compositions were calculated using:

$$x_i = \frac{z_i}{1 + \alpha(K_i - 1)}$$

where the overall compositions  $z_i$  are given, the vapor fraction  $\alpha$  is assigned a trial value, and the equilibrium ratios  $K_i$  depend on the trial outlet temperature through the saturation pressures. Once the liquid-phase compositions were determined, the vapor-phase compositions were calculated from:

$$y_i = K_i x_i$$

Thus, the  $x_i$  and  $y_i$  values were both functions of the guessed outlet temperature and the guessed vapor fraction. These composition values were then used in the ideal-mixture enthalpy expressions for the liquid and vapor phases.

The molar enthalpies for the vapor and liquid phases were evaluated using ideal mixture relations. For the vapor phase, the molar enthalpy is given by:

$$\bar{H}^V = \sum_i y_i H_i^{IG}(T)$$

and for the liquid phase:

$$\bar{H}^L = \sum_i x_i [H_i^{IG}(T) - \Delta H_i^{vap}(T)]$$

These expressions were implemented in two separate cells corresponding to the vapor-phase and liquid-phase molar enthalpies. The ideal gas enthalpies were evaluated using the `HIG()` function, referencing the appropriate component and the trial temperature, while the heats of vaporization were evaluated using the `Hvap()` function, also referencing the appropriate component and temperature. Both functions return values in units of kJ/kmol.

At the inlet, the mixture was determined to be a subcooled liquid, and therefore only the liquid-phase molar enthalpy was required since the vapor fraction is zero. In this case, the liquid compositions  $x_i$  are equal to the overall compositions  $z_i$ , which are known. Evaluating the liquid-phase enthalpy yields:

$$[\bar{H}_{inlet} = -99135.686 \text{ kJ/kmol}]$$

Because the throttling process is isenthalpic, the outlet molar enthalpy must be equal to the inlet molar enthalpy.

At the outlet, the mixture is expected to be in two phases, so the overall molar enthalpy was calculated using the lever rule:

$$\bar{H}_{outlet} = (1 - \alpha)\bar{H}^L + \alpha\bar{H}^V$$

where  $\alpha$  is the vapor fraction. This expression was implemented in a separate cell, referencing the vapor fraction and the previously calculated liquid and vapor phase molar enthalpies.

With the  $K$ -values and molar enthalpy expressions fully defined, the Rachford–Rice equation could then be implemented into the worksheet. This equation, which governs the phase split, is given by:

$$\sum_i \frac{z_i (K_i - 1)}{1 + \alpha (K_i - 1)} = 0$$

This expression was entered into a designated cell, with proper referencing to the calculated  $K$ -values, the known overall compositions  $z_i$ , and the trial vapor fraction  $\alpha$ . Because the  $K$ -values depend on the outlet temperature and the expression itself depends on  $\alpha$ , this formulation allows the equation to be solved numerically within the spreadsheet.

Finally, Excel's Solver tool was used to determine the outlet temperature and vapor fraction simultaneously. The objective function, defined as the Rachford–Rice expression, was set equal to zero by varying the cells corresponding to the outlet temperature and the vapor fraction  $\alpha$ .

In addition to this objective, constraints were applied to ensure physical consistency of the solution. The molar enthalpy at the inlet was required to be equal to the molar enthalpy at the outlet, enforcing the isenthalpic nature of the throttling process. Furthermore, the vapor fraction was constrained such that:

$$0 \leq \alpha \leq 1$$

These conditions ensured that the solution satisfied both the phase equilibrium relations and the energy balance while remaining within physically meaningful limits. The results obtained from Solver indicate that the outlet temperature is  $-45.70^\circ\text{C}$  and the vapor fraction is  $0.3857$ .

| Stream Properties              | Inlet      | Outlet      |
|--------------------------------|------------|-------------|
| Molar Flow, kmol/s             |            |             |
| Mass Flow, kg/s                |            |             |
| Volume Flow, m <sup>3</sup> /s |            |             |
| Temperature, C                 | 0.0000     | -45.6976    |
| Pressure, bar                  | 20.0000    | 1.0000      |
| Vapor Fraction                 | 0.0000     | 0.3857      |
| Molecular Weight, kg/kmol      |            |             |
| Molar Enthalpy, kJ/kmol        | -99135.686 | -99135.686  |
| Vapor Phase                    |            | -39336.664  |
| Liquid Phase                   |            | -136679.977 |
| S2 Two-Phase Calculation       |            | 0.0000      |
| x1                             |            | 0.0390      |
| x2                             |            | 0.5300      |
| x3                             |            | 0.4310      |
| y1                             |            | 0.4565      |
| y2                             |            | 0.4522      |
| y3                             |            | 0.0914      |
| actual:                        |            |             |
| K1                             |            | 11.7147     |
| K2                             |            | 0.8531      |
| K3                             |            | 0.2120      |

Figure 3: Summary of Values After Solver Is Used

|                                  | Outlet |
|----------------------------------|--------|
| Temperature ( $^\circ\text{C}$ ) | -45.70 |
| Vapor Fraction ( $\alpha$ )      | 0.3857 |

Table 2: Temperature and vapor fraction at outlet conditions

## 2 Problem B: Background

In this problem, the solubility of oxygen in water is evaluated at ambient conditions. The system is specified at a temperature of  $25^{\circ}\text{C}$  and a total pressure of 1 atm. The gas phase consists of air, with a dry composition of 21% oxygen and 79% nitrogen, and is at 50% relative humidity. The presence of water vapor in the gas phase reduces the effective partial pressure of oxygen, which must be accounted for in the analysis.

The objective is to determine the concentration of dissolved oxygen in water, expressed in units of mg/L. Because oxygen is a non-condensable gas under these conditions and exists at a temperature above its critical temperature, Henry's law is used to describe the equilibrium between the gas and liquid phases.

Under these conditions, oxygen is present in very small amounts in the liquid phase, while the liquid is composed almost entirely of water. As a result, the system can be treated as a dilute solution, and the equilibrium relationship depends primarily on the partial pressure of oxygen in the gas phase and its corresponding solubility in the liquid phase.

The analysis focuses on determining the correct gas-phase composition accounting for humidity, and then relating this to the liquid-phase concentration of oxygen using appropriate thermodynamic relationships.

### 2.1 Oxygen Solubility in Water

For this problem, Raoult's law cannot be applied, so Henry's law is used to describe the equilibrium behavior of oxygen in water. To begin the calculation, the components H2O, N2, and O2 were entered into the CHECalculator, since these species make up the system being analyzed.

For convenience and future use in the property calculations, the given temperature was converted from degrees Celsius to Kelvin using the `CtoK()` function, and the pressure was converted from atm to bar using the `ATMtoBAR()` function, each referencing the appropriate given values.

From Eq. 5.64 in Dr. C's textbook, the relative humidity is defined as:

$$RH = \frac{P_{H_2O}(T)}{P_{H_2O}^{\text{sat}}(T)}$$

The saturated vapor pressure of water,  $P_{H_2O}^{\text{sat}}(T)$ , was calculated using the `Psat()` function, referencing the water component and the known system temperature. A separate cell was then created for the relative humidity, containing an expression that referenced the trial value of  $P_{H_2O}(25)$  and the calculated value of  $P_{H_2O}^{\text{sat}}(25)$ .

Because the partial pressure of water vapor was not initially known, Goal Seek was used to set the relative humidity cell equal to the given value of 0.5 by changing the cell corresponding to  $P_{H_2O}(25)$ . This provided the water vapor partial pressure consistent with the specified relative humidity of the system.

This Goal Seek procedure resulted in a water vapor partial pressure of approximately  $P_{H_2O}(T) = 0.01607$  bar.

The dry-gas pressure was determined by subtracting the water vapor partial pressure from the total system pressure:

$$P_{\text{dry}} = P_{\text{system}} - P_{H_2O}$$

The partial pressure of oxygen was then calculated by multiplying the dry-gas pressure by the mole fraction of oxygen in dry air:

$$P_{O_2} = 0.21P_{\text{dry}}$$

The partial pressure of oxygen was evaluated to be approximately 0.209 bar.

Using Dalton's law, the gas-phase mole fraction of oxygen was determined from:

$$y_{O_2} = \frac{P_{O_2}}{P_{\text{system}}}$$

Finally, Henry's law was applied to relate the gas-phase composition to the liquid-phase composition of oxygen:

$$y_{O_2}P_{\text{system}} = H_{O_2}x_{O_2}$$

which can be rearranged to give:

$$x_{O_2} = \frac{y_{O_2}P_{\text{system}}}{H_{O_2}} = \frac{P_{O_2}}{H_{O_2}}$$

This expression will be used to calculate the mole fraction of dissolved oxygen in the liquid phase.

Next, the Henry's constant for oxygen was evaluated at the system temperature using the correlation and constants as shown in figure 4:

$$\ln H = A + \frac{B}{T} + C \ln T + DT$$

|          | A           | B        | C        | D           |
|----------|-------------|----------|----------|-------------|
| OXYGEN   | 144.4080745 | -7775.06 | -18.3974 | -0.00944354 |
| NITROGEN | 164.9940745 | -8432.77 | -21.558  | -0.00843624 |
| ARGON    | 169.4780745 | -8137.13 | -23.2547 | 0.00306357  |
| HELIUM   | 117.3450745 | -4397.06 | -16.2365 | 0.00602486  |
| HYDROGEN | 180.0660745 | -6993.51 | -26.3119 | 0.0150431   |
| CO       | 171.7750745 | -8296.75 | 23.3372  | 0           |
| CO2      | 159.8650745 | -8741.55 | -21.669  | 0.00110259  |
| H2S      | 149.5490745 | -8226.5  | -20.2307 | -0.00129405 |

Where the formula for the Henry constant is:

$$\ln H = A + \frac{B}{T} + C \ln T + DT$$

Figure 4: Constants for evaluating Henry's constants

where  $T$  is expressed in Kelvin. After evaluating the right-hand side of this expression, the exponential of the result was taken in order to obtain the Henry's constant in units of bar. Using this procedure, the following value was obtained:

$$H_{O_2} = 44083.38 \text{ bar}$$

The mole fraction of oxygen in the liquid phase was then calculated using Henry's law, since both  $H_{O_2}$  and  $P_{O_2}$  are known. This yields:

$$[x_{O_2} \approx 4.75 \times 10^{-6}; \text{kmol O}_2/\text{kmol total}]$$

The total number of kmoles in the liquid phase is equal to the sum of the kmoles of all species present. However, the mole fractions of nitrogen and oxygen in the liquid phase are extremely small due to their large

Henry’s constants. As a result, their contributions to the total kmols are negligible, and the total can be approximated as the kmols of water alone, which dominates the liquid phase.

To convert the liquid-phase mole fraction of oxygen to the desired units, the value of  $x_{O_2}$  was multiplied by the liquid density obtained from the `RACKET()` function, which returns the density in  $\text{kmol/m}^3$ . This function was evaluated by referencing water as the component, the system temperature of  $25^\circ\text{C}$ , and selecting a solution type of 0 to represent an ideal liquid.

The resulting value was then converted from  $\text{kmol/m}^3$  to  $\text{kmol/L}$  by multiplying by  $\frac{1 \text{ m}^3}{1000 \text{ L}}$ , followed by multiplication with the molecular weight of oxygen using the `MOLW()` function (in  $\text{kg/kmol}$ ), and finally converting from  $\text{kg}$  to  $\text{mg}$  by multiplying by  $10^6$ .

This procedure yields the **final solubility of oxygen in water**:

|                                     |
|-------------------------------------|
| 9.356 mg $\text{O}_2$ /L pure water |
|-------------------------------------|

### 3 Conclusion

In this report, two thermodynamic problems involving phase equilibrium and gas solubility were analyzed using fundamental principles from vapor–liquid equilibrium and mixture thermodynamics.

In Problem A, a multi-component hydrocarbon mixture undergoing a throttling process was examined. By applying Raoult’s law, the Rachford–Rice equation, and an isenthalpic energy balance, the outlet temperature and vapor fraction were determined. The results confirmed that the inlet stream exists as a subcooled liquid, while the outlet stream forms a two-phase mixture due to the pressure reduction. The use of numerical methods, specifically Goal Seek and Solver, allowed for the simultaneous satisfaction of phase equilibrium and energy balance constraints.

In Problem B, the solubility of oxygen in water at ambient conditions was evaluated using Henry’s law. The effect of humidity on the gas-phase composition was accounted for by determining the water vapor partial pressure, which was then used to calculate the oxygen partial pressure. The liquid-phase mole fraction of oxygen was determined and subsequently converted into practical units of  $\text{mg/L}$ . The results demonstrated that oxygen is only sparingly soluble in water, as expected from its large Henry’s constant.

Overall, these problems highlight the application of thermodynamic models to analyze real systems, emphasizing the importance of selecting appropriate equilibrium relationships and utilizing numerical methods to solve coupled nonlinear equations in engineering practice.

Table 3: Summary of Final Results

| Quantity                                      | Condition | Value  |
|-----------------------------------------------|-----------|--------|
| Dew Point Temperature ( $^\circ\text{C}$ )    | Inlet     | 72.80  |
| Dew Point Temperature ( $^\circ\text{C}$ )    | Outlet    | -31.00 |
| Bubble Point Temperature ( $^\circ\text{C}$ ) | Inlet     | 32.75  |
| Bubble Point Temperature ( $^\circ\text{C}$ ) | Outlet    | -73.37 |
| Temperature ( $^\circ\text{C}$ )              | Outlet    | -45.70 |
| Vapor Fraction ( $\alpha$ )                   | Outlet    | 0.3857 |
| Oxygen Solubility in Water ( $\text{mg/L}$ )  | –         | 9.356  |

## References

- [1] M. Caracotsios, *An Introduction to Chemical Engineering Thermodynamics: Theory and Applications*, First Edition, course textbook, Chemical Engineering Department, University of Illinois Chicago, 2026.
- [2] OpenAI, “ChatGPT,” Large Language Model, 2026. [Online]. Available: <https://chat.openai.com>. [Accessed: Apr. 2026].